

CHAPTER 1: REAL NUMBER SYSTEM

1. Historical Motivation: The Crisis of Rigor.

Historically, calculus developed backward: mathematicians used derivatives and integrals for nearly 200 years before establishing a rigorous definition of the real number system.

- *Calculus Without Foundations (1680s–1730s)*: Newton and Leibniz introduced calculus using intuitive notions like “infinitesimals” (dx). The definition and calculations rely on geometric or physical intuitions. While highly effective for physics, these quantities were logically contradictory—being non-zero during division yet zero when vanishing—drawing sharp criticism from philosophers such as Bishop George Berkeley (1734).
- *The ϵ - δ Formalization (1820s–1860s)*: Cauchy and Weierstrass replaced infinitesimals with rigorous limit definitions, which built up the mathematical foundation for calculus. Notably, Weierstrass subsequently constructed pathological counterexamples—such as functions continuous everywhere but differentiable nowhere, demonstrating that *geometric intuition is untrustworthy* in formal analysis.
- *Gaps in the Rational Continuum (1858–1872)*: While preparing introductory calculus lectures in Zürich in 1858, Richard Dedekind realized a fundamental flaw: foundational limit theorems (such as the Intermediate Value Theorem) cannot be proven without assuming the number line has no “holes.” However, the rational numbers $\mathbb{Q}$ contain gaps (e.g., $\sqrt{2} \notin \mathbb{Q}$).

The Core Problem. : $\mathbb{Q}$ is Full of “Holes”. To study analysis, we must guarantee that sequences of numbers getting arbitrarily close together actually converge to a limit within our number system. The rational field $\mathbb{Q}$ fails this criteria. For example, consider $f(x) = x^2 - 2$ on the rational domain $[1, 2] \cap \mathbb{Q}$:

- $f(1) = -1 < 0$ and $f(2) = 2 > 0$.
- $f(x)$ is continuous in $\mathbb{Q}$.
- Yet, there is *no point* $c \in \mathbb{Q}$ where $f(c) = 0$, because $\sqrt{2} \notin \mathbb{Q}$.

In $\mathbb{Q}$, continuous functions can “slip through” the missing points. Therefore, we cannot analyze limits or continuous functions rigorously until we construct $\mathbb{R}$ by filling these rational gaps.

- *The Arithmetization of Analysis (1872)*: Dedekind and Georg Cantor independently published the first rigorous constructions of the real number system $\mathbb{R}$ using purely rational arithmetic, establishing the missing foundation for analysis.

2. Two Constructions of $\mathbb{R}$ (1872).

In 1872, two complementary strategies were formulated to construct $\mathbb{R}$ out of $\mathbb{Q}$ without referencing spatial geometry. We will introduce and discuss in more detail later.

- **Dedekind’s Approach: Order & Cuts**
 - *Core Idea*: A real number is defined by how it partitions or “cuts” the ordered set of rational numbers into two sets.

- *The Cut*: Partition $\mathbb{Q}$ into two non-empty sets (L, R) such that every element in L is strictly less than every element in R .
- *Intuition*: Rational cuts correspond to partitions where R contains a minimum element (or L a maximum). An irrational cut occurs when L has no maximum and R has no minimum—the cut itself is defined to be the real number (e.g., $\sqrt{2}$).
- *Advantage*: Preserves total ordering naturally, making it simple to prove the **Least Upper Bound Property**.
- **Cantor's Approach: Metric & Cauchy Sequences**
 - *Core Idea*: A real number is defined as the limit point of a sequence of rational numbers that cluster together.
 - *Equivalence Classes*: Consider all rational Cauchy sequences (x_n) where $|x_n - x_m| \rightarrow 0$. Two sequences are equivalent if their difference converges to zero.
 - *Intuition*: The sequence $1, 1.4, 1.41, 1.414, \dots$ consists of rational terms, but its limit point is absent in $\mathbb{Q}$. Cantor defines $\sqrt{2}$ as the *equivalence class* of all rational Cauchy sequences approaching that hole.
 - *Advantage*: Based on metric distances, making it straightforward to generalize to higher dimensions and abstract metric spaces.

To prepare for the study of analysis, there are some fundamental questions we need to get better understanding.

Question 1. For $\mathbb{N}$, we cannot list all the numbers and therefore we do not exactly know all the elements in the set and what kind of properties they have. So, can we find a theoretical way to characterize this set?

Question 2. How to understand the extended sets $\mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$ and the operations defined on them?

From our experience, we all know that to get $\mathbb{Q}$ from $\mathbb{Z}$, one simply takes the quotient between two integers $\frac{m}{n}$ with $m, n \in \mathbb{Z}$ and $n \neq 0$. However, this is not rigorous because we even do not know what the operation *division* is before we have the larger set $\mathbb{Q}$. Therefore, we need to give a more theoretical construction for the larger number sets and introduce the operations. Along this process, we will also figure out what kind of properties those sets share and not share. This is important for us to understand the following question.

Question 3. Why do we study calculus on $\mathbb{R}$, not $\mathbb{Q}$?

We will figure out what the properties of $\mathbb{R}$ is crucial for us to build up the framework of calculus. Next, we can extend analysis / calculus to more general topological spaces.

LECTURE 1: NATURAL NUMBERS $\mathbb{N}$

1. The set of natural numbers.

In standard notation, we usually denote the set of *all natural numbers* as

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

But we cannot list all the numbers and therefore we do not exactly know all the elements in the set. So, we would like to give a more rigorous mathematical way to characterize this set. **The mathematical philosophy is to use the properties satisfied by the set to characterize it.** But the properties cannot be too many (it will not be easy to verify), cannot be too few (there are more than one candidates satisfy the properties). For example, to identify a person, we have a few items on the passport (name, data of birth, place of birth,).

The rigorous definition (or description) or the construction of the set of natural numbers, **Nat**, is given by **Peano's Axioms**, named after the Italian mathematician, Peano, who published them in 1889, although he himself attributed them to the German mathematician Dedekind.

It is important to keep in mind that when Peano and others constructed these axioms, their goal was to *provide the fewest axioms that would generate the natural numbers that everyone was familiar with.*

The insight is that this could be done by

- asserting the existence of at least one natural number,
- defining a function, called *successor function*, that can be used to construct the remaining natural numbers.

The **successor function** $S: \text{Nat} \rightarrow \text{Nat}$ is a function that has Nat as its domain. (Intuitively, we should think of $S(x)$ as $x + 1$, but we cannot formally define it this way yet since we do not know what $+$ means! Moreover, we do not know the meaning of 1 here.)

Now, we describe the set Nat by using the following five axioms, which is called **Peano Axioms**:

- N1). $0 \in \text{Nat}$;
(this axiom states existence of a starting element, say 0)
- N2). if $x \in \text{Nat}$, then its successor $S(x) \in \text{Nat}$;
(this axiom states that the existence of the successor function S , i.e. for each element $x \in \text{Nat}$, there is one and only one element $y \in \text{Nat}$ such that $y = S(x)$.)
- N3). 0 is not the successor of any element in Nat ;
(this axiom states that the starting element 0 is not the successor of an element in Nat . Thus for all $x \in \text{Nat}$, $S(x) \neq 0$, i.e. 0 is not a member of range of S . This axiom excludes the possibility of defining Nat to be $\{0\}$)
- N4). if $x, y \in \text{Nat}$ have the same successor, then $x = y$;
(this axiom states that the successor function is an injection; this axiom excludes the possibility of defining Nat to be just $\{0, 1\}$ or a finite set)
- N5). (Axiom of Induction) if A is any subset of Nat such that

- i). $0 \in A$;
 - ii). if $x \in A$, then $S(x) \in A$,
- then $A = \text{Nat}$.

(this axiom basically says that Nat has no non-trivial subset that is “closed” with the successor function.)

Summary: the natural number set Nat is a set of elements with a successor function S defined on it, satisfying the above axioms. Therefore, we formally have that

$$\text{Nat} = \{0, S(0), S(S(0)), S(S(S(0))), \dots\}$$

Of course, to avoid so many nested applications of S , we use the numerals 1, 2, 3 to denote $S(0)$, $S(S(0))$, and $S(S(S(0)))$, respectively, and therefore have

$$\text{Nat} \sim \{0, 1, 2, 3, \dots\} = \mathbb{N}.$$

Therefore, it’s just a symbol that we are using to construct the set. A more concrete example, given

- $\# \in \mathcal{A}$,
- define $S : \mathcal{A} \rightarrow \mathcal{A}$ as $x \rightarrow x\star$

Then, we can get the set

$$\mathcal{A} = \{\#, \#\star, \#\star\star, \#\star\star\star, \#\star\star\star\star, \dots\}.$$

Decimal numbers are, in essence, just a shorthand for this. If we denote

$$0 = \#, 1 = \#\star, 2 = \#\star\star, \dots$$

then, it is clear that

$$\mathcal{A} \sim \{0, 1, 2, 3, \dots\} = \mathbb{N}.$$

Remark 1. The following question can be naturally raised: *does there exist $x \in \mathbb{N}$ with $x \neq 0$ such that x is not the successor of any element of $\mathbb{N}$?*

The answer is NO. Indeed, this follows from N5). Suppose there exists such $x \in \mathbb{N}$ ($x \neq 0$) which is not the successor of any element. We can consider a subset

$$A = \{0, S(0), S(S(0)), \dots\}$$

We can check that A satisfies the two conditions in the Axiom of Induction, therefore, N5) implies $A = \mathbb{N}$. However, $x \notin A$.

2. Axiomatic Arithmetic.

As we know, the importance of $\mathbb{N}$ does not stem from its set-theoretic properties. Rather, $\mathbb{N}$ is of crucial importance because of the arithmetic that we can perform on it. The Peano axioms and the successor function allow us to define **addition** “+” and **multiplication** “.” on $\mathbb{N}$. Indeed, the operations are defined using the following logic:

Successor function - the basic operation;

Addition - repeatedly apply the successor operation;
 Multiplication - repeatedly apply the addition operation.

More precisely,

- We can define **addition** “+” as following:

step 1. For any $a \in \mathbb{N}$, we define

$$a + 0 := a;$$

step 2. For any $a, b \in \mathbb{N}$, we can define recursively

$$a + S(b) := S(a + b).$$

Therefore, we can compute, for example, $a + 1 = a + S(0) = S(a + 0)$.

- We can define **multiplication** “ $\cdot$ ” as following:

step 1. For any $a \in \mathbb{N}$, we define

$$a \cdot 0 := 0;$$

step 2. For any $a, b \in \mathbb{N}$, we can define recursively

$$a \cdot S(b) := a + (a \cdot b).$$

Therefore, we can compute, for example, $a \cdot 1 = a \cdot S(0) = a + (a \cdot 0) = a + 0 = a$.

Proposition 1. *Prove that $(\mathbb{N}, +, \cdot)$ satisfies*

1) *for any $a, b \in \mathbb{N}$, we have*

$$a + b = b + a;$$

2) *for any $a, b, c \in \mathbb{N}$, we have*

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Proof. Exercise! (Apply the definition.)

3. Mathematical induction.

Axiom N5) is the basis of the **induction principle** or **the principle of mathematical induction**.

Let $P_1, P_2, \dots, P_n, \dots$ be a list of statements that may or may not be true. The principle of *mathematical induction* is: if the statements satisfy

i). P_1 is true;

ii). if P_n is true, then P_{n+1} is true,

then, all the statements $P_1, P_2, \dots, P_n, \dots$ are true.

Remark 2. *In general, we call i): the basis for induction and ii): the induction step.*

Why is it true? Consider the set

$$\mathcal{A} = \{n \in \mathbb{N} \mid P_n \text{ is true}\}.$$

It is clear that $\mathcal{A} \subset \mathbb{N}$. The basis for induction and the induction step say that $\mathcal{A}$ satisfies the two conditions in N5), therefore, $\mathcal{A} = \mathbb{N}$.

Example 1. Prove that $1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$ for any $n \in \mathbb{N}$.

Proof: For each $n \in \mathbb{N}$, we view P_n as the statement $1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$. We want to prove that the statements $P_1, P_2, \dots, P_n, \dots$ are all true. Since we can not test all of them one by one, we will use the mathematical induction.

i). basis for induction: P_1 is true, this is because

$$LHS = 1, \quad RHS = \frac{1}{2} \cdot 1 \cdot (1+1) = 1 \implies LHS = RHS$$

ii). induction step: assume P_n is true, that is assuming $1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$, we want to show P_{n+1} is true. Indeed,

$$\begin{aligned} 1 + 2 + 3 + \cdots + n + (n+1) &= \frac{1}{2}n(n+1) + (n+1) = (n+1) \cdot \left(\frac{n}{2} + 1\right) \\ &= \frac{1}{2} \cdot (n+1) \cdot (n+2) \end{aligned}$$

So, P_{n+1} is true.

By the principle of mathematical induction, we know that the statement P_n is true for any $n \in \mathbb{N}$, that is

$$1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$$

for any $n \in \mathbb{N}$. $\square$

Example 2. All numbers of the form $7^n - 2^n$ are divisible by 5.

Proof: Let P_n be the statement that $7^n - 2^n$ are divisible by 5 for $n \in \mathbb{N}$. We want to show the statement P_n is true for any $n \in \mathbb{N}$ by mathematical induction.

i). basis for induction:

$$n = 1 : 7^1 - 2^1 = 7 - 2 = 5$$

which is divisible by 5. Therefore, P_1 is true.

ii). induction step: assume P_n is true, that is $7^n - 2^n$ are divisible by 5, we want to show P_{n+1} is true.

$$n + 1 : 7^{n+1} - 2^{n+1} = 7 \cdot (7^n - 2^n) + 7 \cdot 2^n - 2 \cdot 2^n = 7 \cdot (7^n - 2^n) - 5 \cdot 2^n$$

The second term on the right hand side is obviously divisible by 5, the first term is also divisible by 5 by the assumption that P_n is true. Therefore, P_{n+1} is true.

By the principle of mathematical induction, we know that the statement P_n is true for any $n \in \mathbb{N}$, that is *all numbers of the form $7^n - 2^n$ are divisible by 5*. $\square$

4. Recall some basic math logic.

- 1). Mathematical induction;
- 2). Let A, B and C denote three statements.
 - (i). $A \Rightarrow B$ (A implies B) means *if A is true, then B is true*.
 - (ii). If $A \Rightarrow B$ and $B \Rightarrow C$, then $A \Rightarrow C$.
 - (iii). $A \Leftrightarrow B$ means $A \Rightarrow B$ and $B \Rightarrow A$.
 - (iv). " $A \Rightarrow B$ " $\iff$ "Not $B \Rightarrow$ Not A ".